AI Applications Lecture 18

Parameter Efficient Fine Tuning (PEFT) and LoRA

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[a]I thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction

2. Preliminaries: Mathematical Notations

3. Why parameter-efficient parameter tuning?

4. How do we efficiently tune the parameter? LoRA

5. How do we apply PEFT (LoRA) in practical GenAI?

6. Summary

Background

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Can we modify/tune the AI behavior more efficiently?

Learning Outcomes

At the end of this lecture, you will be able to

- Explain the objective of parameter-efficient fine-tuning.
- Modify the behavior of a NN using low-rank adaptation (LoRA).
- Apply LoRA to a practical generative AI.

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Preliminaries: mathematical notations

Set:

- Sets: $\mathcal{A}$
- Membership: $x \in \mathcal{A}$
- Empty set: $\{\}$
- Roster notation: $\{a, b, c\}$
- Set-builder: $\{x \in \mathcal{A} | P(x)\}$
- Cardinality: $|\mathcal{A}|$
- Real numbers: $\mathbb{R}, \mathbb{R}_{>0}, \mathbb{R}_{\geq 0}$
- Integers: $\mathbb{Z}, \mathbb{Z}_{>0}, \mathbb{Z}_{\geq 0}$
- Integer range: $\{1, 2, ..., k\}$

Function:

- $f : \mathcal{X} \to \mathcal{Y}$: f maps from $\mathcal{X}$ to $\mathcal{Y}$.
- $y = f(x)$: The output of f for input x.

Definition:

- $(\mathrm{LHS}) \coloneqq (\mathrm{RHS})$: Left side is defined by the right side.

Preliminaries: mathematical notations

Sequence: Denoted by $\boldsymbol{a} = (a_0, a_1, \ldots, a_{n-1})$.

- A function $\boldsymbol{a} : \{0, 1, ..., n-1\} \to \mathcal{A}$.
- Length is denoted by $|\boldsymbol{a}|$.

Vector: Denoted by $\boldsymbol{v}$.

- A column of numbers, $\boldsymbol{v} \in \mathbb{R}^n$.
- i-th element is v_i.

Matrix: Denoted by $\boldsymbol{A}$.

- $m \times n$ matrix: $\boldsymbol{A} \in \mathbb{R}^{m,n}$.
- (i, j)-th element is $a_{i,j}$.
- Transpose: $\boldsymbol{A}^\top$.

Tensor: Denoted by $\underline{\boldsymbol{A}}$.

- Simply a multi-dimensional array.
- Vector $\to$ 1st-order, Matrix $\to$ 2nd-order.

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Question: Why do we apply parameter-efficient fine tuning? How effective?

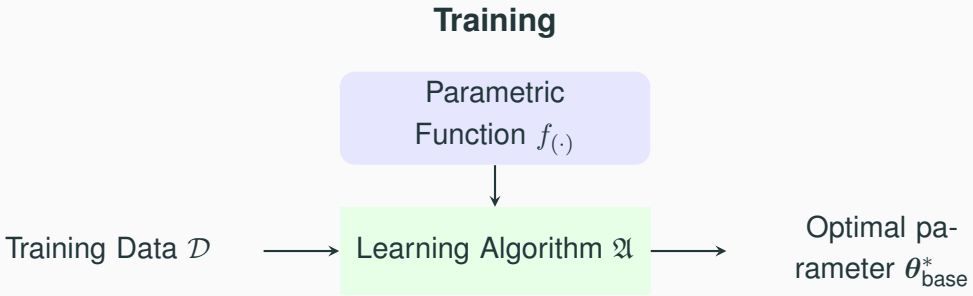
AI as a parametric function: review

Let $\mathcal{X}$ be the input set, $\mathcal{Y}$ the output set, and $\boldsymbol{\theta}_{\text{pretrain}} \in \mathbb{R}^{d_{\text{pretrain}}}$ a parameter vector. A parametric model is a function $f_{\boldsymbol{\theta}_{\text{pretrain}}} : \mathcal{X} \longrightarrow \mathcal{Y}$, mapping $\boldsymbol{x} \longmapsto f_{\boldsymbol{\theta}_{\text{pretrain}}}(\boldsymbol{x})$. For supervised data $D = ((\boldsymbol{x}_i, y_i))_{i=1}^n$ and a loss $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$, full training solves [1]

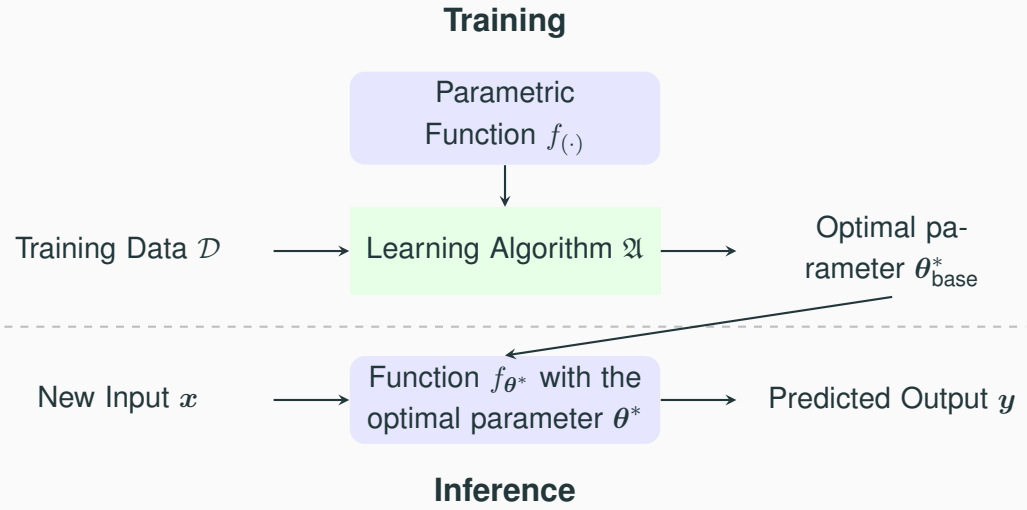
$$\boldsymbol{\theta}^*_{\text{full}} \in \underset{\boldsymbol{\theta}_{\text{full}} \in \mathbb{R}^{d_{\text{pretrain}}}}{\arg\min} \frac{1}{n} \sum_{i=1}^{n} \ell(f_{\boldsymbol{\theta}_{\text{full}}}(\boldsymbol{x}_i), y_i) \,.$$

- **Training**: determine $\boldsymbol{\theta}^*_{\text{full}}$ from D.
- **Inference**: for an unseen $\boldsymbol{x} \in \mathcal{X}$, compute $f_{\boldsymbol{\theta}^*_{\text{full}}}(\boldsymbol{x})$ without changing the parameters.

AI as a parametric function: review



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Parameter-efficient fine-tuning (PEFT): definition

Start from pre-trained, fixed parameters $\boldsymbol{\theta}_{\text{pretrain}} \in \mathbb{R}^{d_{\text{pretrain}}}$.

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From the original parametric function (NN) $f_{(\cdot)}$, PEFT first defines $f^{\text{PEFT}}_{(\cdot,\cdot)} : \mathcal{X} \to \mathcal{Y}$ taking two parameter vectors $\boldsymbol{\theta}_{\text{pretrain}}$ and $\boldsymbol{\theta}_{\text{PEFT}}$.

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Training: We fix $\boldsymbol{\theta}_{\text{pretrain}}$ and tries to find a good $\boldsymbol{\theta}_{\text{PEFT}}$ so $f^{\text{PEFT}}_{\boldsymbol{\theta}_{\text{pretrain}}, \boldsymbol{\theta}_{\text{PEFT}}}$ performs well.

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2. **Reducing required storage**. The pretrained d_{pretrain}-dimensional parameter vector can be shared between tasks. When we have K tasks:
   - Naive fine-tuning requires Kd_{pretrain} scalar parameters.
   - PEFT approach only requires $d_{\text{pretrain}} + Kd_{\text{PEFT}}$ scalar parameters.
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3. **Easy training.** An optimization algorithm updates d_{PEFT} rather than d_{pretrain} scalar parameters. This can reduce optimization cost and increase optimization stability. Thus, training can be done on a small computer and small dataset.

Stable Diffusion 1.5: concrete adapter size

Example (SD 1.5-compatible model and Minimalist Anime Style LoRA)

Consider Stable Diffusion 1.5 as an example. Its (pretrained) parameter counts are $d_{\text{pretrain}} = 1{,}066{,}235{,}307 (\approx 1.07\text{B})$.

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Take Minimalist Anime Style LoRA as an example: Its parameter counts are $d_{\text{PEFT}} = 75{,}497{,}736 (\approx 75\text{M})$. It's just about 7.08% of the pretrained model!

Source: Minimalist Anime Style, `https://civitai.com/models/24833/minimalist-anime-style`; file `anime_minimalist_v1-000020.safetensors`, `https://huggingface.co/hyu8828/Minimalist-Anime/tree/main`; SafeTensors metadata and tensor-shape audit.

Exercise

A pretrained model has 1B parameters; each PEFT adapter has 75M. Using FP16 and 15GB of Google Drive storage, how many tasks can we store with (i) full fine-tuning and (ii) one shared base model plus adapters? Assume 1 GB $= 10^9$ bytes.

Answer: FP16 uses 2 bytes per parameter. Full fine-tuning: $\lfloor (15-2)/(1 \cdot 2) \rfloor = 6$ tasks. PEFT: $\lfloor (15-2)/(0.075 \cdot 2) \rfloor = 86$ tasks.

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- easily revert the modification by “detaching” the parameter difference.
- reduce the data size to store the parameter difference from the original model.
- be achieved by training on a small computer and small dataset.

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What trick enables us to modify the AI effectively with a small number of parameters?

Representative technique: Low rank adaptation (LoRA)

LoRA: a low-rank adaptation

A modern NN architecture has many matrix parameters, e.g., scaled-dot attention, 1x1 convolution, MLP, etc.
The low-rank adaptation (LoRA) [2] replaces a matrix parameter $\boldsymbol{W}_{\text{pretrain}}$ in a NN by

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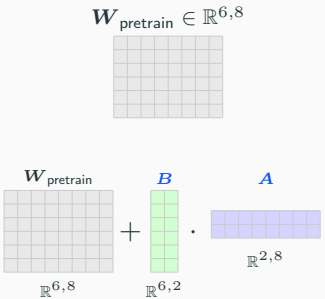
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η can control the modification and is determined at the inference phase.



Source: https://github.com/huggingface/peft/blob/main/src/peft/tuners/lora/layer.py#L2584-L2613.

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- General finetuning replaces $\boldsymbol{W}_{\text{pretrain}} \in \mathbb{R}^{d_{\text{out}}, d_{\text{in}}}$ by $\boldsymbol{W}_{\text{pretrain}} + \boldsymbol{\Delta W}$, where $\boldsymbol{\Delta W} \in \mathbb{R}^{d_{\text{out}}, d_{\text{in}}}$ may take any form.
- LoRA uses $\boldsymbol{A} \in \mathbb{R}^{r, d_{\text{in}}}$ and $\boldsymbol{B} \in \mathbb{R}^{d_{\text{out}}, r}$ with small r. It replaces $\boldsymbol{W}_{\text{pretrain}}$ by $\boldsymbol{W}_{\text{pretrain}} + \Delta \boldsymbol{W}$, where $\Delta \boldsymbol{W} := \frac{\eta}{2} \boldsymbol{B} \boldsymbol{A}$.

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How many scalar parameters do we need to save for each?

- General finetuning: $d_{\text{out}}, d_{\text{in}}$ scalar parameters.
- LoRA: $d_{\text{out}} r + r d_{\text{in}} = r(d_{\text{out}} + d_{\text{in}})$ scalar parameters.

Exercise

For $d_{\text{out}} = 1000$, $d_{\text{in}} = 100$, and $r = 10$, compare the parameter counts of full fine-tuning and LoRA.

Answer: Full: $1000 \cdot 100 = 100{,}000$. LoRA: $10(1000 + 100) = 11{,}000$ (11% of full).

Exercise: which input vector does LoRA change?

Exercise

Let $d_{\text{in}} = 3$, $d_{\text{out}} = 4$, $r = 2$, and $\eta = 1$, $\alpha = r = 2$. Define

$$\boldsymbol{W}_{\text{pretrain}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{4,3}, \quad \boldsymbol{A} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \in \mathbb{R}^{2,3}, \quad \boldsymbol{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix} \in \mathbb{R}^{4,2}.$$

For $\boldsymbol{x}^{(1)} = (1, 2, 1)^\top \in \mathbb{R}^3$ and $\boldsymbol{x}^{(2)} = (-1, 1, 1)^\top \in \mathbb{R}^3$,
compute $\boldsymbol{W}_{\text{pretrain}}\boldsymbol{x}$ and $(\boldsymbol{W}_{\text{pretrain}} + \boldsymbol{B}\boldsymbol{A})\boldsymbol{x}$.

Also, observe when LoRA changes the output.

Exercise: answer and verification

Answer:

$$\boldsymbol{W}_{\text{pretrain}}\boldsymbol{x}^{(1)} = (1,2,1,4)^\top, \quad (\boldsymbol{W}_{\text{pretrain}} + \boldsymbol{BA})\boldsymbol{x}^{(1)} = (6,3,5,15)^\top,$$
$$\boldsymbol{W}_{\text{pretrain}}\boldsymbol{x}^{(2)} = (-1,1,1,1)^\top, \quad (\boldsymbol{W}_{\text{pretrain}} + \boldsymbol{BA})\boldsymbol{x}^{(2)} = (-1,1,1,1)^\top.$$

Since $\boldsymbol{A}\boldsymbol{x}^{(2)} = \boldsymbol{0}$, LoRA does not change the output for $\boldsymbol{x}^{(2)}$.

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In what sense is LoRA's representation limited?

Answer: LoRA can only change the behavior for an input in a low-dimensional subspace.

LoRA cannot change the behavior for all possible inputs

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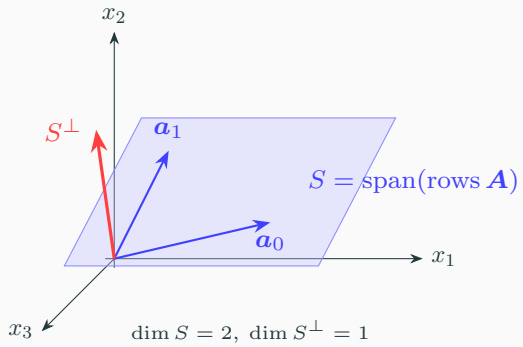
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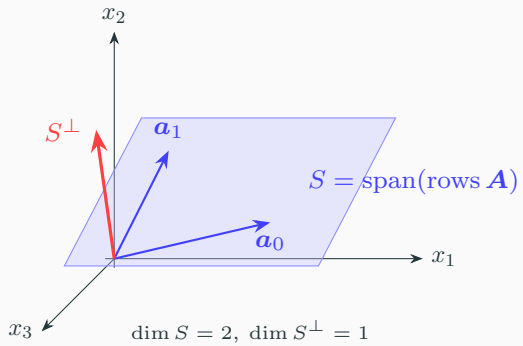
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Define its orthogonal complement $S^\perp := \{\boldsymbol{x} \in \mathbb{R}^{d_{\text{in}}} \mid \boldsymbol{x}^\mathsf{T}\boldsymbol{a} \text{ for all } \boldsymbol{a} \in \text{rows of } \boldsymbol{A}\}$.



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Define its orthogonal complement $S^\perp := \{\boldsymbol{x} \in \mathbb{R}^{d_{\text{in}}} \mid \boldsymbol{x}^\mathsf{T}\boldsymbol{a} \text{ for all } \boldsymbol{a} \in \text{rows of } \boldsymbol{A}\}$.



If $\boldsymbol{x} \in S^\perp$, then $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$. Thus, $\Delta \boldsymbol{W}\boldsymbol{x} = \frac{\eta}{2}\boldsymbol{B}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$.

This means the LoRA cannot change the input in $S^\perp$.

Moreover, $\dim S \leq r$, so $\dim S^\perp \geq d_{\text{in}} - r$. LoRA sacrifices representing changes for inputs in at least $d_{\text{in}} - r$ dimensional subspace.

Matrix rank and the name Low-Rank Adaptation

LoRA replaces $\boldsymbol{W}_{\text{pretrain}}$ by $\boldsymbol{W}_{\text{pretrain}} + \Delta\boldsymbol{W}$, where $\Delta\boldsymbol{W} := \frac{\eta}{2}\boldsymbol{B}\boldsymbol{A}$.

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Specifically, $\mathrm{rank}(\Delta\boldsymbol{W}) \leq r$ in LoRA, if $\eta > 0$.

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Hence, we hope that, after LoRA training, the subspace $S := \mathrm{span}(\text{rows of } \boldsymbol{A})$ covers the important input subspace in application.

In practice, the idea works well. Simple LoRA training (backprop + SGD) can find such an important subspace and change the NN behavior as we want.

Note on implementation: r, α, and η

In the `peft` library, the LoRA is implemented in the following form:

$$\Delta \boldsymbol{W} := \eta \frac{\alpha}{r} \boldsymbol{B}\boldsymbol{A},$$

where $\alpha \in \mathbb{R}$ is a hyperparameter.

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Hence, we use the following form:

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LoRA — Answer

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While we give up a full rank update, it works well practically.

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2. Preliminaries: Mathematical Notations

3. Why parameter-efficient parameter tuning?

4. How do we efficiently tune the parameter? LoRA

5. How do we apply PEFT (LoRA) in practical GenAI?

6. Summary

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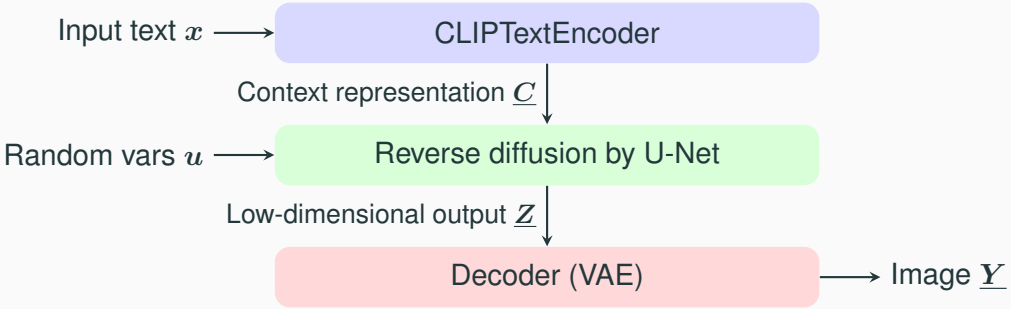
5. How do we apply PEFT (LoRA) in practical GenAI?

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Question: LoRA can apply to matrix parameters in GenAI. Which parts

Question: LoRA can apply to matrix parameters in GenAI. Which parts are the targets of LoRA in a practical GenAI specifically?

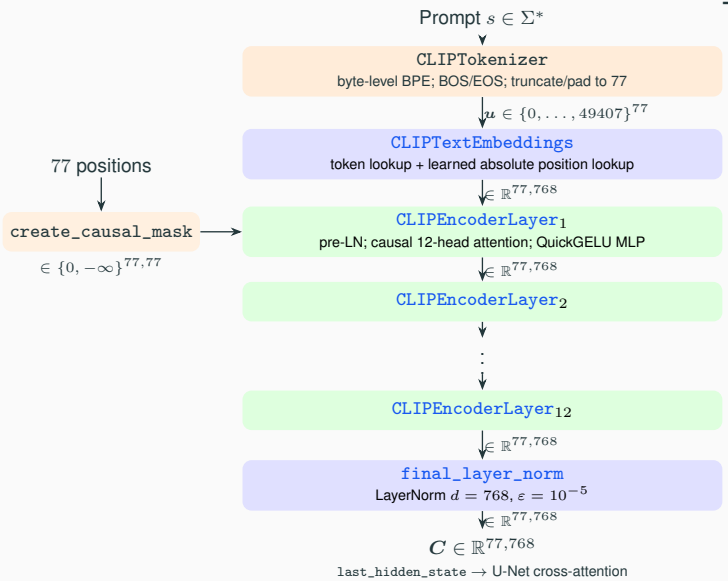
Review of Stable diffusion 1.5



In real PEFT situations, LoRA is applied to

- CLIP text encoder
- U-Net in the reverse diffusion

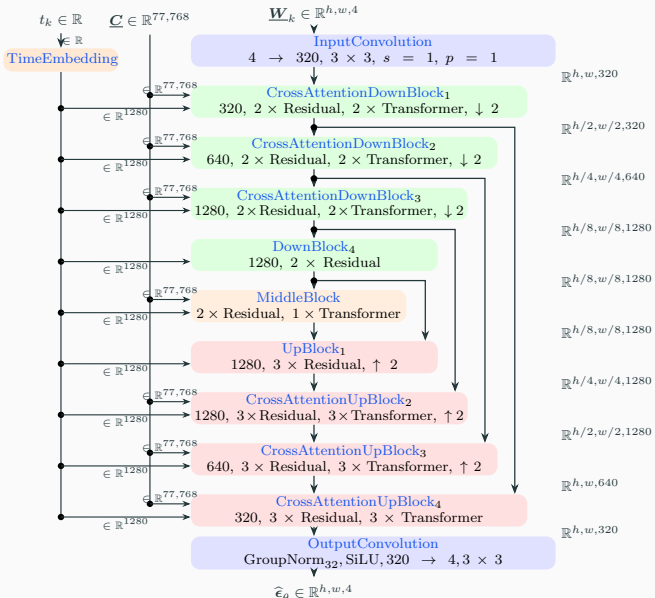
Stable Diffusion 1.5 text encoder: CLIP ViT-L/14 — Whole structure



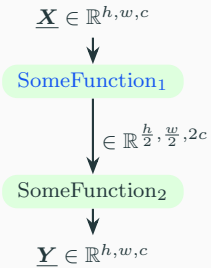
The structure is simple:

- Truncate/padding to 77 tokens
- Embedding
- Attention layers

Stable Diffusion 1.5 U-Net



Legend



Rounded boxes: functions.
Arrows: tensors; the adjacent superscripts give their shapes.
Function in ThisColor indicates it has learnable parameters.

$h = H/8, w = W/8$; image dimensions are divisible by 64.

U-Net outputs predicted noise.

Source: SD 1.5 U-Net configuration; Diffusers UNet2DConditionModel.

Which SD 1.5 layers contain this LoRA?

Take `anime_minimalist_v1-000020.safetensors` as an example.

Network	Target	Modules	$\boldsymbol{A}, \boldsymbol{B}$ entries
CLIP	Self-attention $\boldsymbol{W}_{\mathrm{Q}}, \boldsymbol{W}_{\mathrm{K}}, \boldsymbol{W}_{\mathrm{V}}, \boldsymbol{W}_{\mathrm{O}}$	48	9,437,184
CLIP	MLP `fc1,fc2`	24	11,796,480
U-Net	Self-attention $\boldsymbol{W}_{\mathrm{Q}}, \boldsymbol{W}_{\mathrm{K}}, \boldsymbol{W}_{\mathrm{V}}, \boldsymbol{W}_{\mathrm{O}}$	64	12,779,520
U-Net	Cross-attention $\boldsymbol{W}_{\mathrm{Q}}, \boldsymbol{W}_{\mathrm{K}}, \boldsymbol{W}_{\mathrm{V}}, \boldsymbol{W}_{\mathrm{O}}$	64	12,730,368
U-Net	FFN `net.0.proj,net.2`	32	22,364,160
U-Net	`proj_in,proj_out` (1×1 conv)	32	6,389,760
	Total	264	75,497,472

They are all for token-wise linear operations.

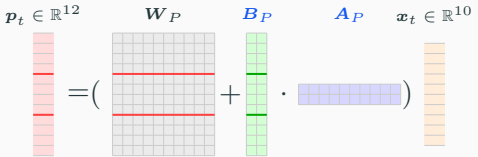
It does **NOT** contain LoRA for the VAE or for U-Net ResNet 3×3 convolutions.

CLIP: LoRA acts on text-token vectors

Let $t \in \{0, \ldots, 76\}$ be a text-token position and $\boldsymbol{x}_t \in \mathbb{R}^{768}$ the CLIP hidden state at one of 12 encoder layers. CLIP has $H_{\text{clip}} = 12$ heads, so $d_h = 768/12 = 64$.

For $P \in \{\mathsf{Q}, \mathsf{K}, \mathsf{V}\}$, define independently

$$\boldsymbol{W}_P \in \mathbb{R}^{768,768},$$
$$\boldsymbol{A}_P \in \mathbb{R}^{128,768}, \quad \boldsymbol{B}_P \in \mathbb{R}^{768,128},$$
$$\boldsymbol{p}_t = (\boldsymbol{W}_P + \tfrac{1}{2}\boldsymbol{B}_P\boldsymbol{A}_P)\boldsymbol{x}_t \in \mathbb{R}^{768}.$$



Split $\boldsymbol{p}_t$ into $\boldsymbol{p}_t^{[h]} \in \mathbb{R}^{64}$, $h \in \{1, \ldots, 12\}$.
Note that $\boldsymbol{p}_t^{[h]} = \boldsymbol{q}_t^{[h]}, \boldsymbol{k}_t^{[h]}, \boldsymbol{v}_t^{[h]}$ for $P = \mathsf{Q}, \mathsf{K}, \mathsf{V}$, respectively.

$\boldsymbol{A}_P$ is shared among 12 heads (within the same projection).

Sources: text encoder config,

CLIP: multi-head attention and its other LoRA targets

For $h \in \{1, \dots, 12\}$ and token positions t, s, the causal attention score and probabilities are

$$a_{t,s}^{[h]} := \frac{(\boldsymbol{q}_t^{[h]})^\top \boldsymbol{k}_s^{[h]}}{\sqrt{64}}, \qquad p_{t,s}^{[h]} := \frac{e^{a_{t,s}^{[h]}}}{\sum_{u=0}^{t} e^{a_{t,u}^{[h]}}} \quad (0 \le s \le t).$$

$$\boldsymbol{o}_t^{[h]} := \sum_{s=0}^{t} p_{t,s}^{[h]} \boldsymbol{v}_s^{[h]} \in \mathbb{R}^{64}, \quad \boldsymbol{o}_t := \begin{bmatrix} \boldsymbol{o}_t^{[1]} \\ \vdots \\ \boldsymbol{o}_t^{[12]} \end{bmatrix} \in \mathbb{R}^{768}.$$

The output projection also has LoRA: $\boldsymbol{y}_t = (\boldsymbol{W}_\mathsf{O} + \frac{1}{2}\boldsymbol{B}_\mathsf{O}\boldsymbol{A}_\mathsf{O})\boldsymbol{o}_t$, where $\boldsymbol{W}_\mathsf{O} \in \mathbb{R}^{768,768}$, $\boldsymbol{A}_\mathsf{O} \in \mathbb{R}^{128,768}$, $\boldsymbol{B}_\mathsf{O} \in \mathbb{R}^{768,128}$.

Also, LoRA usually modifies both MLP linear maps: $\mathbb{R}^{768} \to \mathbb{R}^{3072} \to \mathbb{R}^{768}$.

Sources: https://github.com/huggingface/transformers/blob/main/src/transformers/models/clip/modeling_clip.py#L2555-L2588; https://github.com/huggingface/transformers/blob/main/src/transformers/models/clip/modeling_clip.py#L2590-L2675.

LoRA in U-Net

LoRA is applied to the following places in U-Net:

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- Token-wise fully connected layer
- 1x1 convolution: it is also (essentially) token-wise fully connected layer.

One U-Net Transformer: 12 LoRA modules

An SD 1.5 U-Net contains 16 spatial Transformer blocks: $2 + 2 + 2$ in down blocks, 1 in the middle, $3 + 3 + 3$ in up blocks.

- `proj_in` and `proj_out`: 2 LoRA modules;
- `attn1` (self): $\boldsymbol{W}_{\mathrm{Q}}, \boldsymbol{W}_{\mathrm{K}}, \boldsymbol{W}_{\mathrm{V}}, \boldsymbol{W}_{\mathrm{O}}$, 4;
- `attn2` (cross): $\boldsymbol{W}_{\mathrm{Q}}, \boldsymbol{W}_{\mathrm{K}}, \boldsymbol{W}_{\mathrm{V}}, \boldsymbol{W}_{\mathrm{O}}$, 4;
- `ff.net`: `0.proj` and `2`, 2.

Hence $16 \cdot 12 = 192$ U-Net LoRA modules. The remaining $12 \cdot 6 = 72$ modules modify CLIP.

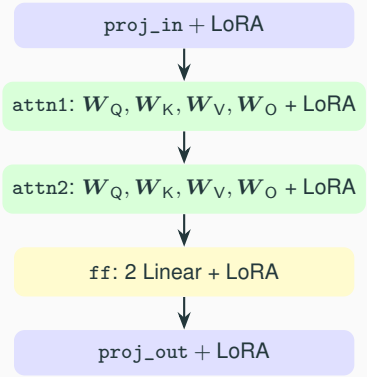


Diagram identifies targets; normalizations and residuals omitted.

Source: file tensor-key audit; https://github.com/CompVis/stable-diffusion/blob/main/ldm/modules/attention.py#L975-L1028;

LoRA in practice — Answer

Question: Which parts are the targets of LoRA in a practical GenAI specifically?

Answer:

LoRA in practice — Answer

Question: Which parts are the targets of LoRA in a practical GenAI specifically?

Answer: In Stable diffusion 1.5, LoRA is applied to token-wise matrix multiplications in the CLIP encoder and U-Net.

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Summary

- PEFT can represent an AI behavior change with a few parameters. It saves storage cost and makes training easier.
- Rank-r LoRA represents the fine-tuning by the product of an r column matrix and an r-row matrix. Small r saves the storage cost, while sacrificing the representability.
- The targets of LoRA in SD 1.5 are token-wise linear operations in CLIP and U-Net, e.g., $\boldsymbol{W}_{\mathrm{Q}}, \boldsymbol{W}_{\mathrm{K}}, \boldsymbol{W}_{\mathrm{V}}, \boldsymbol{W}_{\mathrm{O}}$ in attention layers, token-wise MLP/FFN, 1x1 convolutional layers. LoRA does not apply to VAE or 3x3 convolutional layers.

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[1] Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
Deep Learning.
MIT Press, 2016.

[2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen.
LoRA: Low-rank adaptation of large language models.
In International Conference on Learning Representations, 2022.